Chapter 1. Linear Regression and Gradient Descent

Regression, normal equations, gradient descent, generalization

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1 The Regression Problem

In regression we observe pairs $(\mathbf{x}_i, y_i)$, $i = 1, \dots, n$, with input $\mathbf{x}_i \in \mathbb{R}^k$ and real response y_i, and we want a function that predicts y from $\mathbf{x}$. The simplest choice is a linear function. We motivate the choice of "best line" by the sum of squared errors (SSE): a horizontal benchmark line gives one SSE, rotating it changes the SSE, and plotting SSE against the rotation angle gives a curve that first decreases and then increases. The best line is the minimizer of this curve.

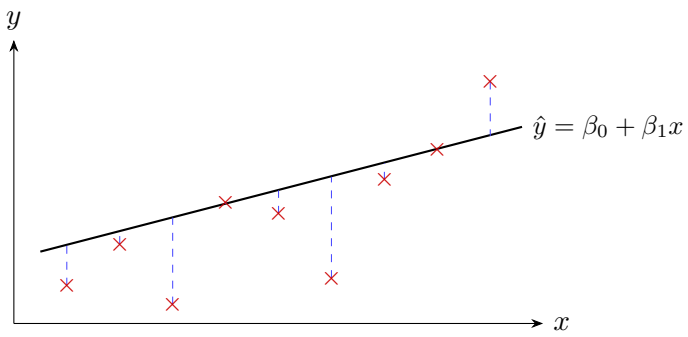
Definition 1.1: linear regression model and design matrix

$$y_i = \beta_0 + \sum_{j=1}^{k} \beta_j x_{ij} + \varepsilon_i, \qquad i = 1, \dots, n.$$

Absorb the intercept with a constant input 1: let $d = k + 1$, $\boldsymbol{\beta} = (\beta_0, \dots, \beta_k)^\top \in \mathbb{R}^d$, and let the **design matrix** $X \in \mathbb{R}^{n \times d}$ have ith row $(1, x_{i1}, \dots, x_{ik})$. Then

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad f(\boldsymbol{\beta}) = \sum_{i=1}^{n} (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 = \|\mathbf{y} - X\boldsymbol{\beta}\|^2 = (\mathbf{y} - X\boldsymbol{\beta})^\top (\mathbf{y} - X\boldsymbol{\beta}).$$

Rows of X index samples; columns index features.



Residuals $y_i - \hat{y}_i$ (dashed) of a fitted line; least squares minimizes the sum of their squares.

2 Matrix Calculus Toolkit

Motivation. Every learning algorithm in the course minimizes a scalar loss over a vector or matrix of parameters. Rather than differentiate coordinate by coordinate, we collect a few identities for gradients of linear and quadratic forms that make such computations one-line.

We use the gradient convention: the gradient of a scalar with respect to a vector is a column vector of the same size.

Theorem 1.2: basic gradients

For $\mathbf{a} \in \mathbb{R}^d$ and $A \in \mathbb{R}^{d \times d}$,

$$\nabla_{\boldsymbol{\beta}}(\mathbf{a}^\top \boldsymbol{\beta}) = \mathbf{a}, \qquad \nabla_{\boldsymbol{\beta}}(\boldsymbol{\beta}^\top A \boldsymbol{\beta}) = (A + A^\top)\boldsymbol{\beta}, \qquad \nabla^2_{\boldsymbol{\beta}}(\boldsymbol{\beta}^\top A \boldsymbol{\beta}) = A + A^\top.$$

If A is symmetric the last two become $2A\boldsymbol{\beta}$ and $2A$.

Proof. $\mathbf{a}^\top \boldsymbol{\beta} = \sum_j a_j \beta_j$, so $\partial / \partial \beta_k = a_k$. For the quadratic form, $\boldsymbol{\beta}^\top A \boldsymbol{\beta} = \sum_{i,j} A_{ij} \beta_i \beta_j$. Differentiating with respect to β_k, the product rule picks up the terms with $i = k$ and those with

$j = k$:

$$\frac{\partial}{\partial \beta_k} \sum_{i,j} A_{ij}\beta_i\beta_j = \sum_j A_{kj}\beta_j + \sum_i A_{ik}\beta_i = (A\boldsymbol{\beta})_k + (A^\top \boldsymbol{\beta})_k.$$

Differentiating once more gives the constant matrix $A + A^\top$. ■

3 Normal Equations

Motivation. Least squares is the one learning problem with a closed-form solution, and it is the benchmark for everything else. Setting the gradient to zero gives the normal equations; their geometry, projection of $\mathbf{y}$ onto the column space, explains what the fit is doing.

Theorem 1.3: least squares solution

$\nabla f(\boldsymbol{\beta}) = 2X^\top(X\boldsymbol{\beta} - \mathbf{y})$ and $\nabla^2 f = 2X^\top X \succeq 0$. Every minimizer solves the **normal equations**

$$X^\top X\boldsymbol{\beta} = X^\top \mathbf{y}.$$

If X has full column rank d, then $X^\top X \succ 0$ and the unique minimizer is $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1}X^\top \mathbf{y}$.

Proof. Expand $f = \mathbf{y}^\top\mathbf{y} - 2\mathbf{y}^\top X\boldsymbol{\beta} + \boldsymbol{\beta}^\top X^\top X\boldsymbol{\beta}$ (the two cross terms are equal scalars). With the toolkit, $\nabla f = -2X^\top\mathbf{y} + 2X^\top X\boldsymbol{\beta}$ and $\nabla^2 f = 2X^\top X$. For every $\mathbf{v}$, $\mathbf{v}^\top X^\top X\mathbf{v} = \|X\mathbf{v}\|^2 \geq 0$, so f is convex and stationary points are global minimizers. If the columns of X are linearly independent, $X\mathbf{v} \neq \mathbf{0}$ whenever $\mathbf{v} \neq \mathbf{0}$; then $X^\top X \succ 0$, it is invertible, f is strictly convex and the minimizer is unique. ■

Remark 1.4: geometry and rank

The normal equations say $X^\top(\mathbf{y} - X\hat{\boldsymbol{\beta}}) = \mathbf{0}$: the residual is orthogonal to every column of X, so $X\hat{\boldsymbol{\beta}}$ is the orthogonal projection of $\mathbf{y}$ onto the column space. Since the first column is $\mathbf{1}_n$, the residuals sum to zero. Having $n > d$ does not by itself guarantee full rank; with dependent columns the minimizer is not unique and the pseudoinverse gives the minimum-norm solution. Numerically one solves the linear system by QR or SVD rather than forming an inverse.

Example 1.5: least squares on three points

Points $(0,1), (1,2), (2,2)$ and model $y = \beta_0 + \beta_1 x$:

$$X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad X^\top X = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \quad X^\top \mathbf{y} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}, \quad (X^\top X)^{-1} = \frac{1}{6}\begin{pmatrix} 5 & -3 \\ -3 & 3 \end{pmatrix}.$$

Hence $\hat{\boldsymbol{\beta}} = \frac{1}{6}(25 - 18, -15 + 18)^\top = (7/6,\ 1/2)^\top$. Predictions are $(7/6, 5/3, 13/6)$, residuals $(-1/6, 1/3, -1/6)$ (sum zero), and SSE $= 1/6$.

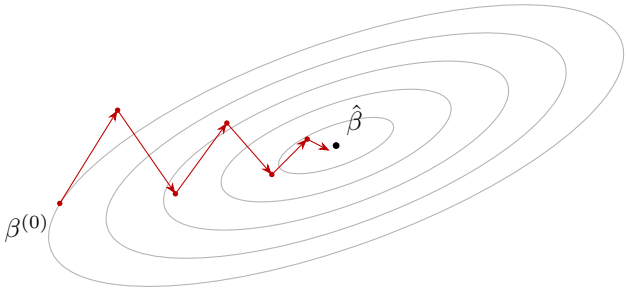
4 Gradient Descent

4.1 The algorithm

Instead of solving the normal equations directly we may iterate

$$\boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t - \eta \nabla f(\boldsymbol{\beta}_t) = \boldsymbol{\beta}_t + 2\eta X^\top (\mathbf{y} - X\boldsymbol{\beta}_t),$$

i.e. componentwise $\beta_j \leftarrow \beta_j + 2\eta \sum_i (y_i - \hat{y}_i) x_{ij}$ with $x_{i0} = 1$. The factor 2 is often absorbed into the learning rate η. Each full-batch step uses all samples. The negative gradient is the direction of steepest descent: for a unit vector $\mathbf{v}$ the first-order change $\nabla f^\top \mathbf{v}$ is minimized, by Cauchy–Schwarz, at $\mathbf{v} = -\nabla f / \|\nabla f\|$.



Gradient descent on the level sets of a quadratic objective: each step moves against the gradient, so an elongated bowl produces a zigzag path.

4.2 Convergence on least squares

Write $J(\boldsymbol{\beta}) = \frac{1}{2}\|X\boldsymbol{\beta} - \mathbf{y}\|^2$ (so $\nabla J = X^\top(X\boldsymbol{\beta} - \mathbf{y})$) and $H = X^\top X$, assumed positive definite.

Theorem 1.6: exact convergence condition

Gradient descent $\boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t - \eta \nabla J(\boldsymbol{\beta}_t)$ converges to $\hat{\boldsymbol{\beta}}$ for every initial point if and only if

$$0 < \eta < \frac{2}{\lambda_{\max}(X^\top X)}.$$

Proof. Error recursion. Since $\nabla J(\hat{\boldsymbol{\beta}}) = \mathbf{0}$, $X^\top \mathbf{y} = H\hat{\boldsymbol{\beta}}$, so $\nabla J(\boldsymbol{\beta}_t) = H(\boldsymbol{\beta}_t - \hat{\boldsymbol{\beta}})$. With $\mathbf{e}_t = \boldsymbol{\beta}_t - \hat{\boldsymbol{\beta}}$,

$$\mathbf{e}_{t+1} = \mathbf{e}_t - \eta H \mathbf{e}_t = (\mathbf{I} - \eta H)\mathbf{e}_t, \qquad \mathbf{e}_t = (\mathbf{I} - \eta H)^t \mathbf{e}_0.$$

Diagonalize. H is symmetric, so $H = Q\Lambda Q^\top$ with Q orthogonal and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_d)$, $\lambda_i > 0$. Using $\mathbf{I} = QQ^\top$, $\mathbf{I} - \eta H = Q(\mathbf{I} - \eta\Lambda)Q^\top$. Let $\mathbf{c}_t = Q^\top \mathbf{e}_t$. Multiplying the recursion by $Q^\top$ and using $Q^\top Q = \mathbf{I}$ gives $\mathbf{c}_{t+1} = (\mathbf{I} - \eta\Lambda)\mathbf{c}_t$, a diagonal recursion:

$$c_{t,i} = (1 - \eta\lambda_i)^t \, c_{0,i}.$$

Because $\|\mathbf{c}_t\| = \|\mathbf{e}_t\|$, $\mathbf{e}_t \to \mathbf{0}$ iff every coordinate $c_{t,i} \to 0$.

Sufficiency. If $|1 - \eta\lambda_i| < 1$ for all i, every coordinate tends to zero for every $\mathbf{e}_0$.

Necessity. If $|1 - \eta\lambda_i| \geq 1$ for some i, start from $\mathbf{e}_0 = \mathbf{q}_i$ (the ith eigenvector). Then $\mathbf{e}_t = (1 - \eta\lambda_i)^t \mathbf{q}_i$ and $\|\mathbf{e}_t\| = |1 - \eta\lambda_i|^t \geq 1$, so this initial point does not converge.

Solve. $|1 - \eta\lambda_i| < 1 \iff 0 < \eta\lambda_i < 2 \iff 0 < \eta < 2/\lambda_i$. Requiring this for all i is equivalent to $0 < \eta < 2/\lambda_{\max}$. ∎

Remark 1.7: conditioning

The contraction factor of direction i is $|1-\eta\lambda_i|$. The rate $\max_i |1-\eta\lambda_i|$ is minimized by $\eta^\star = 2/(\lambda_{\max}+\lambda_{\min})$, giving rate $(\kappa-1)/(\kappa+1)$ with condition number $\kappa = \lambda_{\max}/\lambda_{\min}$. Elongated (ill-conditioned) level sets make gradient descent slow; this is revisited in Chapter 6. At $\eta = 2/\lambda_{\max}$ the steepest direction oscillates with factor -1 and never decays, so the inequality is strict. In Example 1.5, $\lambda(X^\top X) = 4 \pm \sqrt{10}$ and the condition is $0 < \eta < (4-\sqrt{10})/3 \approx 0.279$.

5 Polynomial Features and Generalization

5.1 Feature maps

For scalar input take $\boldsymbol{\phi}_M(x) = (1, x, x^2, \dots, x^M)^\top$ and $y(x) = \sum_{j=0}^{M} \beta_j x^j$. The model is nonlinear in x but linear in the coefficients, so all of the above applies with the (Vandermonde-type) design matrix whose rows are $\boldsymbol{\phi}_M(x_i)^\top$.

5.2 Training error versus generalization

Because a degree-M model contains every lower-degree model, the minimal training SSE cannot increase as M increases. Validation error can nevertheless increase: a high-degree fit follows noise and oscillates between data points (overfitting), while too low a degree cannot capture the trend (underfitting). Performance is measured by the root mean squared error

$$\mathrm{MSE} = \frac{1}{n}\mathrm{SSE}, \qquad E_{\mathrm{RMS}} = \sqrt{\mathrm{MSE}}.$$

Remark 1.8: roles of the data splits

Training data choose the coefficients; validation data choose the degree and the regularization strength; a held-out test set estimates final performance. In typical learning curves the gap between training and validation RMS grows with M for $M \geq 5$, and degree 9 with $n = 10$ overfits badly while $n = 100$ points support the same degree much better. "About ten points per parameter" is a heuristic, not a theorem. Regularization (Chapter 3) is the second remedy.

6 Proof Supplement: Uniqueness and the Pseudoinverse

Motivation. When $X^\top X$ is singular the normal equations have many solutions. The pseudoinverse picks the one of minimum norm, which is also the solution gradient descent converges to from zero, an early example of implicit regularization.

Theorem 1.9: set of minimizers

The minimizers of $\|\mathbf{y} - X\boldsymbol{\beta}\|^2$ are exactly the solutions of $X^\top X\boldsymbol{\beta} = X^\top \mathbf{y}$; this system is always consistent, and the solution set is $\hat{\boldsymbol{\beta}}_0 + \ker X$ for any particular solution $\hat{\boldsymbol{\beta}}_0$. Among them, $X^+\mathbf{y}$ has minimal Euclidean norm.

Proof. Consistency: $X^\top \mathbf{y}$ lies in the column space of $X^\top$, which equals the column space of $X^\top X$ (both have orthogonal complement $\ker X$, since $X^\top X\mathbf{v} = \mathbf{0} \Rightarrow \|X\mathbf{v}\|^2 = 0$). Two solutions differ by $\mathbf{v}$ with $X^\top X\mathbf{v} = \mathbf{0}$, i.e. $\mathbf{v} \in \ker X$, and give the same fitted values. Write $X = U\Sigma V^\top$ (thin

SVD with r positive singular values). Every solution is $V_r \Sigma_r^{-1} U_r^\top \mathbf{y} + \mathbf{v}$ with $\mathbf{v} \perp \text{range}(V_r)$; by Pythagoras its norm is minimized at $\mathbf{v} = \mathbf{0}$, which is $X^+ \mathbf{y}$. ■